AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Google Datacenter - Douglas County, GA -- station KFTY, America/New_York
Building	OSM 975064004 Switch Data Center
Facade gap	not applicable -- one building, no second facade
Weather record	43,666 real hourly records from KFTY
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-12-10 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 23 hours with 2 mode changes. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	10 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	7.220	18.0	8.330	4.571
01	FREE	yes	-	7.770	18.0	8.890	4.246
02	FREE	yes	-	7.780	18.0	8.890	3.330
03	FREE	yes	-	7.780	18.0	8.890	2.594

04	FREE	yes	-	8.330	18.0	8.890	2.658
05	FREE	yes	-	8.330	18.0	9.440	2.264
06	FREE	yes	-	8.330	18.0	9.440	1.929
07	FREE	yes	-	10.000	18.0	9.440	2.217
08	FREE	yes	-	12.220	18.0	9.440	2.467
09	FREE	yes	-	16.110	18.0	10.000	4.101
10	mechanical	no	dry-bulb	18.890	18.0	10.000	4.797
11	mechanical	no	dry-bulb	22.220	18.0	10.560	5.927
12	mechanical	no	dry-bulb	23.880	18.0	11.670	6.387
14	mechanical	no	dry-bulb	22.220	18.0	12.220	5.417
15	mechanical	no	dry-bulb	18.890	18.0	12.220	3.797
16	mechanical	yes	switch budget	16.670	18.0	12.220	2.956
17	mechanical	yes	switch budget	14.440	18.0	12.220	2.825
18	mechanical	yes	switch budget	12.790	18.0	12.220	2.690
19	mechanical	yes	switch budget	12.230	18.0	12.780	3.148
20	mechanical	yes	switch budget	11.110	18.0	12.780	3.264
21	mechanical	no	dew point	10.560	18.0	12.780	3.986
22	mechanical	no	dew point	10.550	18.0	13.330	4.617
23	mechanical	no	dew point	10.560	18.0	13.890	4.566

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.220 C, which is 10.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.571 C, and the margin is measured, not chosen: 4.571 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.770 C, which is 10.230 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.246 C, and the margin is measured, not chosen: 4.246 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.780 C, which is 10.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.330 C, and the margin is measured, not chosen: 3.330 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.780 C, which is 10.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.594 C, and the margin is measured, not chosen: 2.594 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 9.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.658 C, and the margin is measured, not chosen: 2.658 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 9.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.264 C, and the margin is measured, not chosen: 2.264 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 9.670 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.928 C, and the margin is measured, not chosen: 1.928 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 8.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.217 C, and the margin is measured, not chosen: 2.217 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.220 C, which is 5.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.467 C, and the margin is measured, not chosen: 2.467 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.110 C, which is 1.890 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.101 C, and the margin is measured, not chosen: 4.101 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 18.0 C limit, so it fails by 4.220 C. It would take a limit 4.220 C higher, or a bound 4.220 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.880 C against a 18.0 C limit, so it fails by 5.880 C. It would take a limit 5.880 C higher, or a bound 5.880 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 18.0 C limit, so it fails by 4.220 C. It would take a limit 4.220 C higher, or a bound 4.220 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.670 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.440 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.790 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.230 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 10.560 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 10.550 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

23:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 10.560 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

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